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Reflection on AI-Powered Product Description Generator

For this assignment, I created an AI-powered Product Description Generator to explore how businesses can use generative AI in retail and e-commerce. The application uses Hugging Face Transformers (Option B) with a small Qwen language model, which allows it to run in Google Colab without a paid API key. Using Python and Flask, I developed a simple web interface where users can enter a product name, provide keywords, and choose between four writing tones, including Friendly, Professional, Fun, and Luxury. The main goal was to create an interactive application that could generate personalized product descriptions for different customers while exploring how language models can support online shopping experiences.

During development, testing different products, such as reusable water bottles and wireless Bluetooth headphones, helped reveal both the strengths and limitations of the model. One of the biggest challenges involved the AI generating product details that the user never provided, including claims about premium materials or additional performance benefits. Even after several prompt adjustments, the model occasionally continued adding unsupported information. To reduce these mistakes, I implemented an automated screening system that checks for certain questionable claims, retries the generation process, and provides a simpler description when necessary. However, further testing showed that the screening system could not catch every inaccurate statement. Another challenge involved making the writing tones noticeably different, especially Luxury and Professional, which initially produced very similar results. A possible improvement would be connecting the application to a database of verified product information, allowing the model to create descriptions using accurate details instead of relying only on user input and prompt instructions.

In conlusion, this assignment provided valuable experience with LLMs, Flask, and prompt engineering while demonstrating how AI can contribute to personalized shopping experiences. The results showed that even small language models can generate creative descriptions, although they sometimes struggle to follow specific instructions consistently. From an ethical perspective, businesses need to review AI-generated product descriptions carefully because misleading claims could influence customers' purchasing decisions and damage their trust. Most importantly, this project helped me understand that AI can make content creation more efficient, but human judgment still plays an important role in ensuring accuracy and reliability.